

ECON1010

Introductory Microeconomics

*Complete Exam Revision Notes
Topics 1 – 10 | All Concepts Explained*

Scarcity & Choice

Opportunity Cost

Supply & Demand

Elasticity

Perfect Competition

Monopoly

Game Theory

Externalities

Public Goods

Asymmetric Info

Price Discrimination

Market Failure

Topic 1A

Thinking Like an Economist

Lectures 1 & 2 — Scarcity, Choice, Trade-offs & the Logic of Rational Decisions

Economics: The Science of Choice Under Scarcity

Economics is NOT just about money or markets — it is a way of thinking about any situation where resources are limited and choices must be made.

The Scarcity Principle

- Resources (time, money, land, labour) are ALWAYS limited
- Having more of one thing means having less of another
- This forces every individual, firm, and government to make TRADE-OFFS
- Even time is scarce — choosing to study means not watching Netflix

Why Does This Matter?

- Because scarcity is universal, economics applies EVERYWHERE
- Not just businesses — also health, relationships, government, sport
- Every decision involves allocating scarce resources
- Economics gives us a rigorous toolkit to analyse ANY decision

KEY INSIGHT: Economics is the science of how rational agents make choices when faced with scarcity. The fundamental question is always: 'Is it worth it?'

- Positive Economics: describes what IS (e.g. 'A minimum wage of \$25 increases unemployment by X%')
- Normative Economics: prescribes what SHOULD be (e.g. 'The minimum wage should be raised') — involves value judgements

The Cost-Benefit Principle & Marginal Analysis

The Golden Rule of Rational Decision-Making

Take an action if, and only if, the **EXTRA BENEFIT** $\geq$ the **EXTRA COST** → Think at the margin, not the average!

Marginal = Incremental

- 'Marginal' means 'one more unit'
- Should a firm produce one more unit?
- → Only if $MR \geq MC$
- Should you study one more hour?
- → Only if extra mark gained $\geq$ opportunity cost of that hour
- Ignore averages; focus on the **NEXT** unit

Sunk Costs — Ignore Them!

- Sunk cost = already spent; cannot be recovered
- Rational decision: treat sunk costs as **IRRELEVANT**
- Example: Bought a \$100 non-refundable concert ticket
- If you feel sick on the night, the ticket cost is sunk
- Decision = only: is going worth the marginal cost tonight?
- Sunk cost fallacy = acting on irrecoverable past spending

Economic Surplus

- Economic Surplus = Total Benefits – Total Costs
- Includes **ALL** costs: explicit + implicit (opportunity)
- A positive surplus → action is worthwhile
- A negative surplus → action destroys value
- Rational agents act to **MAXIMISE** their economic surplus
- Markets maximise **TOTAL** economic surplus at equilibrium

EXAM TIP: 'Marginal' is not the same as 'small' — it means the **NEXT** unit, whatever size that is.

Worked Example

- Firm sells widgets at \$20. MC of 101st unit = \$18. Produce it? YES (benefit \$20 > cost \$18). MC of 102nd = \$22? NO.

Opportunity Cost — The True Cost of Every Choice

Every choice has a hidden price tag: what you give up

Opportunity Cost = the value of the NEXT BEST ALTERNATIVE foregone. It is NOT just money — it includes time, effort, and any implicit benefit missed.

Why Opportunity Cost is Tricky

- It includes **IMPLICIT** costs — things with no price tag
- Going to university: tuition is explicit; foregone salary is implicit
- A business owner's salary they could earn elsewhere is an implicit cost
- Economists count both; accountants often only count explicit costs
- Always ask: 'What is the **BEST** alternative I am giving up?'

Worked Examples

- You earn \$60,000 as a teacher. Could earn \$80,000 as an engineer → implicit OC = \$20,000/yr
- You study 3hrs for ECON instead of MGMT → OC = value of 3hrs of MGMT study
- Govt builds a road on land worth \$2M → OC = \$2M of best alternative use (e.g. park)
- Holding \$10,000 cash → OC = interest/return if invested elsewhere

Explicit Costs

Actual money paid out-of-pocket
e.g. wages, rent, materials

Implicit Costs

Value of resources **YOU** own and use
e.g. your own time, your own building

Opportunity Cost

= Explicit + Implicit
= Total of **EVERYTHING** foregone

Production Possibilities Curve (PPC)

Visualising Scarcity & Trade-offs — The PPC

What the PPC Shows

- All **MAXIMUM** combinations of 2 goods an economy can produce using all resources efficiently
- **ON** the curve → Productively Efficient (no waste)
- **INSIDE** the curve → Inefficient (idle/wasted resources, e.g. unemployment)
- **OUTSIDE** the curve → Currently **UNATTAINABLE**
- The **SLOPE** of the PPC = opportunity cost of producing more of one good
- The PPC shifts **OUTWARD** with: technological progress, more resources, better education
- A PPC shift = economic **GROWTH**

Points ON the PPC show the trade-off; the gap between inside and the curve = efficiency gains available.

Why is the PPC Bowed Outward?

- Law of Increasing Opportunity Cost: Resources are **NOT** equally suited to all uses.
- First, use most-suited resources → low OC. Eventually must use poorly-suited ones → OC rises.

The 'Low-Hanging Fruit Principle'

- Start with the easiest/cheapest resources first.
- As you produce **MORE** of one good, you must divert increasingly unsuitable resources → each extra unit costs more.

Straight-Line PPC Exception

- If OC is **CONSTANT** (resources equally good at both tasks), the PPC is a straight line.
- This is rarely realistic but simplifies many exam questions.

Trade & the PPC

- Trade allows a country to **CONSUME** outside its PPC by specialising (see comparative advantage).

WHY Countries (and People) Specialise & Trade

The key insight of comparative advantage: BOTH parties can gain from trade even if one is better at EVERYTHING. Trade is about relative, not absolute, efficiency.

Absolute Advantage

- Person/country A has an absolute advantage in good X if it can produce MORE of X per unit of input than B.
- Example: Australia produces 10 wheat OR 5 wine per worker-hour; NZ produces 6 wheat OR 4 wine.
- Australia has absolute advantage in BOTH goods.

Comparative Advantage

- Person/country A has a comparative advantage if it produces good X at a LOWER OPPORTUNITY COST.
- Australia: OC of wheat = 0.5 wine. NZ: OC of wheat = 0.67 wine → Australia has CA in wheat.
- NZ: OC of wine = 1.5 wheat. Australia: OC of wine = 2 wheat → NZ has CA in wine.
- Each should SPECIALISE in their CA good and trade.

	Absolute Advantage	Comparative Advantage
Definition	Produce MORE with same inputs	Produce at LOWER opportunity cost
Basis for Trade?	No — not sufficient alone	YES — the true basis for trade
Can both parties have it?	Yes, in different goods	Each party has CA in at least one good
Exam relevance	Less important	CRITICAL — always tested

Topic 2

Demand and Supply

Lectures 3 & 4 — How Markets Work, Equilibrium & Government Intervention

Demand: What Consumers Are Willing & Able to Buy at Each Price

Law of Demand: As price RISES, quantity demanded FALLS (inverse/negative relationship). WHY? Substitution effect + income effect.

The Demand Curve

- Downward sloping: as price falls, Qd rises
- Each point = maximum willingness to pay (WTP) for that unit
- The demand curve IS the marginal benefit curve
- Area under demand curve = total consumer benefit
- Consumer Surplus = WTP – Price Paid (benefit consumers capture)

WHY Does Demand Slope Down?

- Substitution effect: when good X gets expensive, switch to substitutes
- Income effect: higher price = lower real purchasing power → buy less
- Diminishing marginal utility: each extra unit gives less satisfaction → willing to pay less for it
- These three forces combine to give the downward slope

DEMAND SHIFTERS — what moves the ENTIRE curve (not just a point on it)

Income

Normal good: $\uparrow$ income $\rightarrow \uparrow$ demand
Inferior good: $\uparrow$ income $\rightarrow \downarrow$ demand

Prices of Related Goods

Substitutes: $\uparrow$ price of Pepsi $\rightarrow \uparrow$ demand for Coke
Complements: $\uparrow$ price of petrol $\rightarrow \downarrow$ demand for

Tastes/Preferences

Fashion, advertising, health reports change willingness to pay

Population/Demographics

More people = more demand; ageing population shifts demand profile

Expectations

Expect price to rise next month $\rightarrow$ buy more NOW (anticipate)

Supply: What Firms Are Willing & Able to Sell at Each Price

Law of Supply: As price RISES, quantity supplied RISES (positive relationship). WHY? Higher price covers higher marginal costs AND attracts new suppliers.

The Supply Curve

- Upward sloping: as price rises, Q_s rises
- Each point = minimum price firm needs to supply that unit (= MC)
- Supply curve IS the marginal cost curve (above min AVC)
- Producer Surplus = Price Received – Minimum WTA (benefit producers capture)
- Profit incentive drives firms to supply more at higher prices

WHY Does Supply Slope Up?

- Law of Diminishing Returns: adding more variable inputs to fixed capital → MC rises
- Higher prices needed to cover rising MC of extra units
- Higher prices also attract new firms into market → total market Q_s rises
- Opportunity cost: existing resources pulled from other uses cost more at margin

SUPPLY SHIFTERS — what moves the ENTIRE curve

Input Costs

↑ wages or raw materials → ↑ MC → ↓ supply
(curve shifts left)

Technology

Better tech → produce more cheaply → ↑ supply
(curve shifts right)

Govt Policy

Tax → ↑ cost → ↓ supply. Subsidy → ↓ cost → ↑ supply

of Sellers

More firms enter → market supply increases

Expectations

Expect future price to rise → withhold supply now

How Markets Clear — and What Happens When They Don't

Equilibrium

- Where $Q_d = Q_s \rightarrow$ the 'market clearing price'
- No tendency to change — the market 'rests' here
- Total economic surplus (CS + PS) is MAXIMISED
- At P^* $\rightarrow$ all mutually beneficial trades occur
- Above P^* : surplus $\rightarrow$ price falls.
Below P^* : shortage $\rightarrow$ price rises

Deadweight Loss (DWL)

- Loss of total surplus when market is NOT at equilibrium
- Represents trades that WOULD have been mutually beneficial but DO NOT occur
- Caused by: taxes, subsidies, price controls, monopoly power
- $DWL =$ social cost of market distortion
- Shown as a triangle on S&D diagram between old & new quantity

Consumer & Producer Surplus

- Consumer Surplus (CS): $WTP - \text{Price paid} = \text{consumer's gain}$
- Producer Surplus (PS): $\text{Price received} - \text{Min WTA} = \text{producer's gain}$
- Total Surplus = $CS + PS =$ the 'economic pie'
- Equilibrium maximises this pie — any intervention shrinks it
- A tax transfers some CS & PS to government but creates DWL

Type	Definition	Condition to Bind	Effect	Real World Example
Price Ceiling	Legal MAXIMUM price	Must be set BELOW equilibrium	SHORTAGE ($Q_d > Q_s$), black markets, quality falls	Rent control
Price Floor	Legal MINIMUM price	Must be set ABOVE equilibrium	SURPLUS/excess supply ($Q_s > Q_d$)	Minimum wage, farm price floors

Topic 3

Elasticity

Lecture 5 — Measuring Sensitivity: Price, Income & Cross-Price Elasticity

How Sensitive is Quantity Demanded to Price Changes?

PED = % change in Quantity Demanded ÷ % change in Price Always negative (law of demand) — we usually take the absolute value |PED|.

Perfectly Inelastic |PED| = 0

Qty doesn't change at all with price. Vertical demand curve. E.g. insulin for a diabetic.

Inelastic $0 < |PED| < 1$

Qty changes by LESS % than price. Necessities, few substitutes, short run. Raise price → revenue rises.

Unit Elastic |PED| = 1

Qty changes by SAME % as price. Revenue is maximised HERE.

Elastic |PED| > 1

Qty changes by MORE % than price. Luxuries, many substitutes, long run. Raise price → revenue falls.

Perfectly Elastic |PED| = ∞

Any price rise → qty drops to zero. Horizontal demand curve. E.g. a perfectly competitive firm.

Determinants of Price Elasticity — WHY are some goods more elastic?

Availability of Substitutes

More substitutes → more elastic. If Pepsi rises, easy switch to Coke → demand elastic.

Proportion of Budget

Large share of income → more elastic (pay attention to price). Small share → inelastic.

Necessity vs Luxury

Necessities (bread, medicine) → inelastic. Luxuries (holidays, jewellery) → elastic.

Time Horizon

Longer time → more elastic. People find alternatives, adjust habits. Short run → inelastic.

The Revenue Rule — Critical Exam Application

Total Revenue = Price × Quantity. Whether raising price increases or decreases TR depends ENTIRELY on elasticity.

Elasticity	Price Rises → TR?	Price Falls → TR?	Why?
Inelastic ($ PED < 1$)	TR RISES ↑	TR FALLS ↓	% price change > % Qd change; price effect dominates
Unit Elastic ($ PED = 1$)	TR unchanged	TR unchanged	% price change exactly offsets % Qd change
Elastic ($ PED > 1$)	TR FALLS ↓	TR RISES ↑	% Qd change > % price change; quantity effect dominates

Income Elasticity (YED)

- $YED = \% \Delta Qd \div \% \Delta Income$
- $YED > 0$ → Normal good (buy more as income rises)
- $YED > 1$ → Luxury (e.g. overseas travel)
- $0 < YED < 1$ → Necessity (e.g. bread)
- $YED < 0$ → Inferior good (buy LESS as income rises, e.g. homebrand noodles)

Cross-Price Elasticity (XED)

- $XED = \% \Delta Qd(A) \div \% \Delta Price(B)$
- $XED > 0$ → Substitutes (Pepsi & Coke): ↑ Pepsi price → ↑ Coke demand
- $XED < 0$ → Complements (petrol & cars): ↑ petrol price → ↓ car demand
- $XED \approx 0$ → Unrelated goods
- Magnitude indicates STRENGTH of relationship

Price Elasticity of Supply (PES)

- $PES = \% \Delta Qs \div \% \Delta Price$
- Always positive (law of supply)
- $PES > 1$ → Elastic supply (easily ramp production)
- $PES < 1$ → Inelastic supply (hard to expand quickly)
- Determinants: time, spare capacity, ease of storage, input flexibility

Topic 4

Perfectly Competitive Supply

Lecture 6 — Costs, Profit Maximisation & the Competitive Firm

The Ideally Competitive Market — Features & Cost Structures

Features of Perfect Competition

- Many small buyers AND sellers (no market power)
- Homogeneous / identical products
- Free entry AND exit in the long run
- Perfect information (everyone knows all prices)
- Firms are PRICE TAKERS: accept market price, can't influence it

Short-Run Cost Concepts

- Total Fixed Cost (TFC): does NOT change with output (rent, loan repayments)
- Total Variable Cost (TVC): changes with output (materials, casual labour)
- Total Cost (TC) = TFC + TVC
- Average Fixed Cost (AFC) = TFC/Q
→ always falls as Q rises

• $Average\ Variable\ Cost\ (AVC) = \frac{TVC}{Q}$

Marginal Cost (MC)

- MC = change in TC from producing ONE more unit
- MC eventually RISES due to Law of Diminishing Returns
- MC curve is U-shaped (falls then rises)
- MC intersects AVC and ATC at their MINIMUM points
- WHY? When $MC < ATC$, producing more lowers average. When $MC >$

ATC , producing more raises average.

KEY RELATIONSHIP: MC cuts both AVC and ATC at their minimums. This is a mathematical necessity, not a coincidence.

U-shaped: $ATC = AFC + AVC$

Min = MC curve above min AVC

Law of Diminishing Marginal Returns — The Reason MC Eventually Rises

- Definition: As you add more of a VARIABLE input (e.g. labour) to a FIXED input (e.g. factory), the extra output (marginal product) eventually falls.
- Example: 1 worker in a kitchen produces 20 meals/hr. 2 workers produce 35 meals (MP=15). 3 workers produce 45 meals (MP=10). 4 workers = 50 meals (MP=5). Kitchen is getting crowded — workers get in each other's way.
- Because each extra worker contributes LESS output, you need MORE workers per extra unit of output → cost per extra unit (MC) rises.
- This is a short-run phenomenon — in the long run ALL inputs are variable.

When Should a Firm Produce? How Much? Should It Shut Down?

PROFIT MAXIMISATION RULE: Produce where $MR = MC$. For a perfectly competitive firm, $MR = P$ (price), so the rule is: Produce where $P = MC$.

Why Produce Where $P = MC$?

- If $P > MC$ for next unit → selling it adds more revenue than cost → PROFIT rises → produce it
- If $P < MC$ for next unit → producing it costs more than it earns → PROFIT falls → DON'T produce it
- Where $P = MC$ exactly → no incentive to expand OR contract → profit is maximised
- This rule applies in ALL market structures; just MR differs

from P in perfect competition

The Shutdown Decision (Short Run)

- Fixed costs are SUNK in the short run — paid regardless of output
- Relevant question: does operating cover VARIABLE costs?
- Operate if: $P \geq AVC$ (price covers variable costs; firm reduces its loss)
- Shut down if: $P < AVC$ (operating makes the loss WORSE than shutdown)
- Shutdown point = minimum AVC
- If $P = AVC$ exactly, firm is indifferent (loss = TFC either way)

Short-Run Profit Outcomes

- Economic Profit > 0 : $P > ATC$ at profit-maximising Q → earns above-normal returns
- Normal Profit (Economic Profit = 0): $P = ATC$ → covers ALL costs including opportunity cost
- Economic Loss: $P < ATC$ → losing money but may still operate if $P \geq AVC$
- Profit = $(P - ATC) \times Q$ [can be negative = loss]

A loss in short run does NOT necessarily mean shutdown

Long-Run Equilibrium

- Free entry/exit ensures only NORMAL profits in the long run
- If economic profit > 0 → new firms enter → supply $\uparrow$ → price $\downarrow$ → profit squeezed
- If economic loss → firms exit → supply $\downarrow$ → price $\uparrow$ → losses eliminated
- Long-run equilibrium: $P = \min ATC = MC$ → both productively AND allocatively efficient
- The market 'self-corrects' through the invisible hand mechanism

Topic 5

The Quest for Profit

Lecture 7 — Accounting vs Economic Profit, Normal Profit & Market Entry/Exit

Accounting vs Economic Profit — A Critical Distinction

Why Economists and Accountants 'See' Profit Differently

The difference is **OPPORTUNITY COSTS**. Economists deduct what you **COULD HAVE** earned elsewhere; accountants only deduct what you **actually** paid.

Accounting Profit

- = Total Revenue – Explicit Costs
- Explicit costs = out-of-pocket cash expenses
(wages paid, rent paid, materials purchased)
- DOES NOT include owner's foregone salary
- DOES NOT include opportunity cost of own capital
- Used for tax purposes and financial reporting
- A positive accounting profit doesn't mean it's economically worthwhile!

Economic Profit

- = Total Revenue – Explicit Costs – Implicit Costs
- Implicit costs = opportunity costs of owned resources
(owner's own labour at market rate, own capital at market return)
- Economic profit > 0 → earning **MORE** than best alternative
- Economic profit < 0 → would do **BETTER** elsewhere
- Economic profit = 0 → **NORMAL PROFIT** (exactly covers all opportunity costs)

Normal Profit Explained

- Normal profit ≠ zero accounting profit
- Normal profit = economic profit of **ZERO**
- Means: all resources earning exactly their opportunity cost
- The firm is doing just as well as it could in its **NEXT BEST** alternative
- This is the long-run outcome in competitive markets
- Positive econ. profit → signal for **ENTRY**; negative → **EXIT**

WORKED EXAMPLE: Revenue \$200k. Rent paid \$30k. Staff \$80k. Owner foregoes \$60k salary. Accounting profit = \$90k. Economic profit = \$30k.

The 'Invisible Hand' & Profit Signalling

- Economic profits > 0 signal: 'This industry is doing better than alternatives → resources should flow **HERE**' → new firms enter → supply increases → prices fall → profits normalise.
- Economic losses signal: 'Resources better used elsewhere' → firms exit → supply decreases → prices rise → losses eliminated.
- This self-correcting mechanism (Adam Smith's 'invisible hand') allocates resources efficiently **WITHOUT** central planning.

Topic 6

Monopoly & Monopolistic Competition

Lecture 8 — Market Power, $MR < P$, Deadweight Loss & Long-Run Outcomes

When One Firm Controls the Entire Market

A monopoly is the **ONLY** seller of a good with no close substitutes. It faces the entire downward-sloping market demand — so to sell more, it must lower price for **ALL** units.

Features of Monopoly

- Single seller, unique product
- High barriers to entry: legal (patents, licences), natural (high fixed costs), resource control
- Price **MAKER**: sets its own price by choosing quantity
- Faces downward-sloping demand (unlike price-taker firms)
- Examples: utility companies, patented drugs, NBN infrastructure

WHY $MR < P$ — The Key Insight

- To sell one more unit, monopoly must lower price on **ALL** existing units
- Gain = price of new unit. Loss = price cut on all previous units
- Net $MR < P$ always (except the very first unit)
- MR curve has **TWICE** the slope of the linear demand curve
- $MR = 0$ at midpoint of demand curve (unit elastic point)
- Below midpoint: $MR < 0 \rightarrow$ selling more actually **DECREASES** revenue

Profit Maximisation

- Still maximise where $MR = MC$
- But then **PRICE** is read off the **DEMAND** curve (not MR)
- $P > MR = MC \rightarrow$ monopoly earns economic profit
- Produces **LESS** than socially optimal quantity
- Charges **MORE** than perfectly competitive price
- Economic profit can persist **LONG TERM** due to barriers to entry

Inefficiency & Deadweight Loss

- Competitive outcome: $P = MC$ (allocatively efficient)
- Monopoly outcome: $P > MC \rightarrow$ some consumers willing to pay $> MC$ but **DON'T** get the good
- These 'lost' trades = **DEADWEIGHT LOSS** (social welfare reduction)
- Monopoly extracts surplus **FROM** consumers and converts some to profit, but creates DWL
- Government responses: price regulation, antitrust/competition law, nationalisation

Natural Monopoly

- Occurs when one firm can supply entire market at **LOWER** cost than two firms
- Average cost falls over the entire range of market demand (economies of scale)
- Examples: water pipes, electricity grid, railway tracks
- Breaking up = inefficient (duplicated fixed infrastructure)
- Government solution: regulate price (e.g. $P = ATC$ for breakeven, or $P = MC +$ subsidy)

Many Firms, Differentiated Products — The Real World of Most Markets

Monopolistic competition combines elements of BOTH perfect competition (many firms, free entry) AND monopoly (product differentiation gives some pricing power).

Features

- Many sellers (like perfect competition)
- DIFFERENTIATED products: not identical, but close substitutes (brands, location, quality)
- Free entry and exit in the long run
- Each firm faces downward-sloping demand for its OWN variety
- Examples: cafes, restaurants, clothing brands, hairdressers, law firms

Short Run vs Long Run

- SHORT RUN: Differentiation gives market power → can set $P > MC$ → earn economic profits
- LONG RUN: Profits attract new entrants with similar products → demand for each firm's product falls (shifts left)
- Entry continues until each firm earns only NORMAL PROFIT (economic profit = 0)
- Long-run equilibrium: $P = ATC$ but $P > MC$ (some DWL remains due to differentiation)
- Firms operate with EXCESS CAPACITY (not at min ATC) — price is paid for variety

Feature	Perfect Competition	Monopolistic Competition	Monopoly
# of Firms	Many	Many	One
Products	Identical	Differentiated	Unique
Entry/Exit	Free	Free	Blocked
Price vs MC	$P = MC$ (long run)	$P > MC$	$P \gg MC$
LR Economic Profit	Zero (normal)	Zero (normal)	Positive (if barriers hold)
DWL?	No	Yes (small)	Yes (larger)

Topic 7

Imperfect Competition: Strategy

Lecture 9 — Price Discrimination, Oligopoly & Game Theory

Extracting Consumer Surplus Through Strategic Pricing

Price discrimination = charging DIFFERENT prices to DIFFERENT consumers (or in different quantities) for the SAME good, NOT based on cost differences.

1st Degree (Perfect)

Charge every consumer their EXACT maximum WTP.
Captures ALL consumer surplus → converts to producer profit.
Produces at socially optimal Q ($MR=MC=P$ for each consumer).
 $DWL = 0$ but consumer gets nothing.
Example: negotiate price with each individual buyer (rare).
Requires perfect info about each consumer.

2nd Degree (Quantity)

Charge different prices for DIFFERENT quantities / 'blocks' of consumption.
Consumers self-select into price tiers based on usage.
Examples: electricity tariffs, mobile data plans, bulk discounts.
'Block pricing': first 100kWh at \$0.30, next 200kWh at \$0.22.
Works without knowing individual WTP.

3rd Degree (Group)

Segment market into GROUPS with different demand elasticities; charge different prices per group.
Examples: student/senior discounts, business vs economy airfares, cinema vs Netflix.
High elasticity group: set lower price. Low elasticity group: set higher price.
Requires ability to prevent resale (arbitrage) between groups.

CONDITIONS NEEDED: market power + ability to identify groups + ability to prevent resale between groups

Strategic Interdependence — When Firms Must Think About Each Other

Oligopoly Features

- Few large firms dominate the market (e.g. 2-5 major players)
- High barriers to entry (economies of scale, brand, capital requirements)
- STRATEGIC INTERDEPENDENCE: each firm's profit depends on rivals' decisions
- Products may be homogeneous (steel) or differentiated (cars, airlines)
- Examples: supermarkets (Coles/Woolworths), airlines, banks, smartphones

Game Theory Basics

- Game theory analyses strategic decisions where outcomes depend on ALL players' choices
- A 'game' has: players, strategies, payoffs
- Nash Equilibrium: outcome where NO player can improve by changing strategy ALONE
- Players assume rivals act rationally (in their own self-interest)
- Applied to: pricing, advertising, R&D, international trade policy

The Prisoner's Dilemma — The Core Insight

- Two players each choose independently. Acting in self-interest leads to an outcome WORSE for BOTH than if they had cooperated.

PAYOFF MATRIX (Firm A rows, Firm B columns) — Payoff = (A's profit, B's profit)

	B: Cooperate (Low price)	B: Defect (High price)
A: Cooperate (Low price)	(\$3, \$3) — Both OK	(\$1, \$4) — A suffers, B wins
A: Defect (High price)	(\$4, \$1) — A wins, B suffers	(\$2, \$2) ← NASH EQUILIBRIUM (both defect)

LESSON: Rational self-interest → mutual defection (\$2,\$2). Cooperation would give (\$3,\$3) but is unstable without enforcement. Cartels face this problem!

Topic 8

Externalities & Property Rights

Lecture 10 — Market Failure, Social Costs/Benefits & Policy Solutions

Externalities — When Markets Get It Wrong

Costs & Benefits That Spill Over to Third Parties Not in the Transaction

An externality is a cost OR benefit of a transaction that falls on someone **OUTSIDE** the transaction. Markets ignore these → market **FAILURE** (too much or too little produced).

Negative Externality (External Cost)

- Occurs when production/consumption **HARMS** third parties
- Producer's Private MC < Social MC (social cost = private + external)
- Market overproduces: $Q_{\text{market}} > Q_{\text{optimal}}$
- Examples: pollution, noise, secondhand smoke, traffic congestion
- WHY does market overproduce? Firm ignores costs imposed on others → treats cost as zero
- Solution: Pigouvian TAX = external cost → raises private MC to social MC

Positive Externality (External Benefit)

- Occurs when production/consumption **BENEFITS** third parties
- Consumer's Private MB < Social MB (social benefit = private + external)
- Market underproduces: $Q_{\text{market}} < Q_{\text{optimal}}$
- Examples: education, vaccination, R&D, beekeeping (pollination), renovating your home
- WHY does market underproduce? Consumer ignores benefits flowing to others → WTP understated
- Solution: SUBSIDY = external benefit → raises effective demand/supply to social optimal

Policy Solutions to Externalities

Pigouvian Tax

Tax = marginal external cost. Added to negative externalities. Raises private cost to match social cost. Example: carbon tax, cigarette excise.

Subsidy

Payment = marginal external benefit. Added to positive externalities. Lowers private cost or raises revenue. Example: university subsidies, vaccination rebates.

Coase Theorem

If property rights are clearly defined AND transaction costs are low, private bargaining achieves efficient outcome **WITHOUT** government. Parties negotiate and internalise the externality themselves.

Regulation

Command-and-control: set standards (e.g. emission limits). Simpler but less flexible than taxes. May not achieve cost-effective reduction.

COASE CAVEAT: In reality, transaction costs are often **HIGH** (many parties, free-rider, info problems) → government intervention needed.

Topic 9

Public Goods

Lecture 11 — Non-Rivalry, Non-Excludability, Free-Rider Problem & Optimal Provision

Classifying Goods by Rivalry & Excludability

RIVALRY: Does my consuming the good reduce the amount available for others?
EXCLUDABILITY: Can someone be prevented from consuming the good if they don't pay?

	EXCLUDABLE (can exclude non-payers)	NON-EXCLUDABLE (cannot exclude)
RIVALROUS (my use reduces yours)	PRIVATE GOODS • Bread, clothing, cars, apples • Markets work well → efficient allocation • Normal supply & demand applies • Excludable + Rivalrous = ideal market good	COMMON-POOL RESOURCES (CPRs) • Fish in ocean, forest timber, underground water, shared grazing land • Non-excludable → cannot stop others using • Rivalrous → depletion is real • 'Tragedy of the Commons': overuse → depletion • Requires regulation or property rights
NON-RIVALROUS (my use doesn't reduce yours)	CLUB GOODS (Artificially Excludable) • Toll roads, pay TV, private parks, streaming services, software • Non-rivalrous: one more user costs nothing • But excludable: charge entry fee • Market CAN provide but may under-provide • Congestion can make it rivalrous at high usage	PUBLIC GOODS • National defence, street lighting, public fireworks, lighthouses, free-to-air TV, public knowledge • Non-rivalrous + Non-excludable • FREE-RIDER PROBLEM → market under-provides • Government provides via taxation • Key challenge: determining optimal quantity

EXAM: Know all 4 categories with examples. The 2x2 grid is a favourite exam question.

WHY Markets Under-Provide Public Goods — and How to Fix It

The Free-Rider Problem

- Because non-payers CANNOT be excluded from public goods, rational individuals have incentive to NOT pay and 'free-ride' on others' contributions
- If everyone free-rides → the good is not provided at all (market failure)
- Even if people WANT the good, they won't pay voluntarily
- Examples: everyone benefits from national defence; incentive not to fund it privately
- Solution: COMPULSORY taxation — government provides and forces contribution
- Note: some quasi-public goods (charities, community groups) can be private but still under-provided

Finding the Optimal Quantity of a Public Good

- For PRIVATE goods: horizontal summation of demand curves (each person buys their own quantity)
- For PUBLIC GOODS: VERTICAL summation of demand curves (everyone consumes SAME quantity)
- WHY vertical? Each person's WTP adds up for the SAME unit (non-rival → we all enjoy it simultaneously)
- Social Marginal Benefit (SMB) = sum of all individual MBs at each quantity
- Optimal Q^* where $SMB = MC$ of providing the good
- Government should provide Q^* and fund it through taxation

KEY DISTINCTION: Private goods → ADD quantities (each person gets their own). Public goods → ADD willingness-to-pay values (everyone shares the same unit).

Worked Example: Vertical Summation

- 3 people in a town consider installing a street light ($MC = \$150$)
- Person A: $WTP = \$80$. Person B: $WTP = \$50$. Person C: $WTP = \$40$
- Individual WTPs: $\$80 < \150 , $\$50 < \150 , $\$40 < \150 → nobody pays alone
- SOCIAL $WTP = \$80 + \$50 + \$40 = \$170 > \$150 MC$ → WORTH PROVIDING!
- Market fails (no one pays alone). Government provides → efficient outcome.

Government Provision Challenges

- Information problem: hard to know each person's true WTP (people understate to free-ride)
- Political problem: majority voting may not reflect true preferences
- Bureaucracy: government may not produce at minimum cost
- Solution approaches: surveys, revealed preference, cost-benefit analysis
- Privatisation (where possible): toll roads convert public goods to club goods

Topic 10

Economics of Information

Lecture 12 — Asymmetric Information, Adverse Selection, Moral Hazard & Signalling

Asymmetric Information — When One Side Knows More

Perfect Competition Assumes Perfect Info — Reality Doesn't Deliver It

Asymmetric information: one party in a transaction has BETTER information than the other. This causes markets to fail in predictable ways.

Adverse Selection — Pre-Contract Problem

- Occurs BEFORE the transaction: the party with private info self-selects in ways that harm the other party
- Insurance market example: people who KNOW they are high-risk buy insurance at average rates; healthy people opt out → pool becomes riskier → premiums rise → more healthy people leave → 'death spiral'
- Used car market ('Lemons'): seller knows car quality; buyer doesn't → buyer assumes average quality → willing to pay average price → good cars withdrawn (sellers won't accept avg price for good car) → only lemons remain
- Labour market: job applicants know their own ability; employers don't → hard to set wages

Moral Hazard — Post-Contract Problem

- Occurs AFTER the transaction: once insured/contracted, party changes BEHAVIOUR because they don't bear full consequences
- Insurance: once insured, people take MORE risks (drive faster, don't install smoke alarm, eat badly)
- Employment: once hired with job security, worker may reduce effort (shirking)
- Banking: 'Too big to fail' → banks take more risk knowing govt will bail them out
- WHY? The cost of risky behaviour is borne by others (insurer, employer, taxpayer)
- Solutions: deductibles, co-payments, monitoring, performance pay, bonding

The Principal-Agent Problem

- Principal: the party who hires (employer, shareholder, patient)
- Agent: the party who acts on their behalf (employee, manager, doctor)
- Problem: agent's interests may differ from principal's, AND agent's actions are hard to monitor
- Examples: shareholder (principal) vs CEO (agent); patient vs doctor; government vs bureaucrat
- Solutions: incentive pay (commissions, shares), monitoring, reputation mechanisms, contracts

Signalling — The Costly-to-Fake Principle

- High-quality providers can signal quality credibly IF the signal is TOO COSTLY for low-quality to mimic
- Education as signal: may not increase productivity but signals pre-existing ability to employers
- New car warranty: only a quality manufacturer can afford to offer it → credible signal
- Designer labels, professional licences, money-back guarantees, university degrees
- Screening: the uninformed party induces the informed to REVEAL their type through self-selection (e.g. job trial, insurance deductible choices)

COSTLY-TO-FAKE PRINCIPLE: A signal is credible ONLY if it would be too costly or impossible for a low-quality provider to replicate it.

ECON1010 Exam Cheat Sheet

The Most Tested Concepts — Know These Cold

Opportunity Cost

ALWAYS the next-best alternative foregone. Includes implicit costs. Appears in EVERY topic.

Elasticity & Revenue

Inelastic $\rightarrow \uparrow P$ raises TR. Elastic $\rightarrow \uparrow P$ lowers TR. Revenue max at unit elastic.

Externalities

Negative $\rightarrow$ overproduce $\rightarrow$ Pigouvian tax. Positive $\rightarrow$ underproduce $\rightarrow$ subsidy. Coase if low transaction costs.

Adverse Selection

Pre-contract info asymmetry. High-risk crowd out low-risk. Lemons problem. Fix: screening, mandatory insurance.

Accounting vs Econ Profit

Econ profit = Accounting profit - implicit (opportunity) costs. Normal profit = econ profit of 0.

Price Discrimination

1st=perfect (each WTP). 2nd=quantity blocks. 3rd=group segments. Need: market power + no resale.

P = MC Rule

Profit max for competitive firm. For monopoly: set Q where $MR=MC$, then read P off demand.

DWL

Any distortion (tax, monopoly, price control) creates DWL = lost mutually beneficial trades.

Public Goods

Non-rival + non-excludable $\rightarrow$ free-rider $\rightarrow$ government provides. Optimal Q: vertical sum of MB = MC.

Moral Hazard

Post-contract behaviour change. Fix: monitoring, deductibles, incentive pay.

Comparative Advantage

Lower opportunity cost, NOT more output. Basis for trade. BOTH parties gain.

Nash Equilibrium

No player can improve by changing strategy ALONE. Prisoner's Dilemma = rational individual choice $\rightarrow$ bad joint outcome.